

Reliable and Efficient Image Processing and Deep Machine Learning for large-scale Digital Image Retrieval

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Abstract— Retrieving relevant photos from a database by analysing their differences and similarities is an essential part of machine vision. Numerous applications exist in areas such as storage, object recognition, localisation, and recognition with limited data. In this study, we take a look at the issue of image retrieval, in which a group of pictures is retrieved from a huge database in order to find ones that are most like the one the user has submitted (the "query image"). This thesis investigates the origins of the issue, as well as its present-day manifestations and potential solutions. To the best of our knowledge, however, there is currently no accessible generic text removal method that can be used to any font, script, language, or shape to remove all or user-specified text sections. The difficulties of multi-lingual and curved text identification and inpainting are carried over into the development of a general text eraser for real-world settings. To the best of our knowledge, however, there is currently no accessible generic text removal method that can be used to any font, script, language, or shape to remove all or user-specified text sections. Creating a general text eraser for use in real-world scenarios is difficult since it takes on the difficulties of both multi-lingual text detection and curved text inpainting. By adhering to these standards, our method significantly outperforms the state of the art on four benchmark datasets. This includes more complicated algorithms with auxiliary components. Furthermore, we offer a qualitative evaluation of our trained representation, which shows that despite its small size, it is able to gather information from highly localised and discriminative regions, much like an implicit attention mechanism.

Keywords—Computer Vision, Machine Learning, Deep learning, Image retrieval.

I. INTRODUCTION

Some of the earliest issues in machine vision, content-based image retrieval (CBIR) and image similarity analysis (ISA) have several uses in fields like image archiving, object recognition, and information retrieval. The proliferation of internet-connected devices and the exponential increase in accessible image data have brought renewed focus to these issues, and recent advances in Deep Neural Network (DNN) based algorithms have resulted in novel approaches to image searching. In this thesis, we examine both traditional and cutting-edge techniques and compare and contrast their

performance under realistic testing conditions. Both CBIR and ISA suffer from the same fundamental issue. Both an image database and a query image are available to us. Both of them need to search through the database for pictures that are similar to the query picture. We call this process "picture retrieval" in the paper.

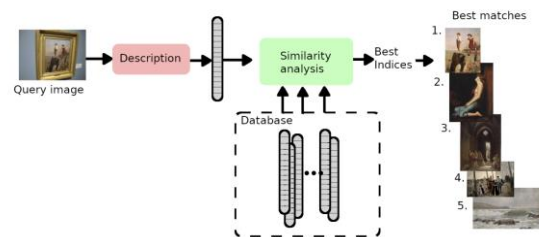


Fig. 1 Process of Image Retrieval

The criteria for what constitutes a successful image match vary widely from one context to the next [1]. The primary goal of content-based image retrieval, or CBIR, is to find photos that depict the same concepts. When the number of distinguishable classes exceeds the capacity of classification networks, it is preferable to convert an annotated image collection into a k-Nearest Neighbors (kNN) classifier, with node distances representing image similarities. This has applications in a variety of recognition tasks. In this context, one-shot learning is analogous to picture retrieval issues.

It is possible, and perhaps likely, that the meaning of "matching images" will change when transferred from one setting to another [1]. Finding similar images is the main focus of content-based image retrieval (CBIR). Related applications include converting an annotated image set into a k-Nearest Neighbors (kNN) classifier with node distances being image similarities when the number of recognisable classes gets too big for classification networks. In this setting, there is a lot of overlap between picture retrieval problems and one-shot learning.

II. BACKGROUND

With this strategy, we query an image database for a medical image using all of its potential features. The final feature set is a compilation of all the features and is compared to the feature set of a database. Some of the features that can be calculated for use in image statistical feature extraction are contrast, correlation, energy, and homogeneity. The values of the pixels can be used to determine the inverse difference motion of the pixels. Area, circumference, and circularity can all be determined, as can the centroid of various regions and the centroid midpoint of a picture.

Simple architecture of a content-based picture retrieval system is shown in Figure 2. The test set is a collection of photos that serve as the system's input. Image descriptors, such as colour, texture, and shape, are retrieved from the test set and saved as feature vectors for later use. The photos in the database can be compared visually using a feature database formed from the images' respective image descriptors. The test set's feature vector that corresponds to the database pictures' feature vector is retrieved for the similarity comparison. Since the volume of medical images is constantly increasing, content-based image retrieval can make a phenomenal contribution in medical applications for diagnosis and research. This necessitates a reliable system for archiving and retrieving visual content.

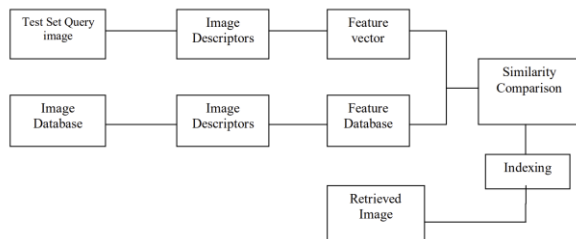


Fig. 2 Block diagram for the Content based Image Retrieval process

This research focuses on creating a practical system for medical image retrieval that is content-based and uses low-level attributes like colour, shape, and texture. Despite the fact that we will only be able to develop a minimally viable prototype, the concept may be adequate to fulfil the needs for a more robust implementation of medical picture retrieval. Just as important as finding the right image in a database for a given query is how long it takes to find it. If your content-based picture retrieval system takes too long to execute a query but returns correct results rapidly, it is of limited use. An interactive, content-based image retrieval system that could swiftly and accurately address the unique needs of the medical community would be of immense use. Also, a content-based image retrieval system should make it simple for its users to enter queries and acquire results; a cumbersome process for doing either would turn off the system's users. Therefore, it is preferable to have a graphical user interface (GUI) built into the content-based picture retrieval system to enable entry of query and get the appropriate output. The query image's quality is a major factor in the results that are obtained. Making a retrieval image that is perceptually comparable to the target images to be recovered is the true challenge when dealing with medical images.

III. METHODOLOGY

Color, shape, texture, and spatial arrangement are just some of the visual contents of a picture that can be used by an image retrieval system for identification and indexing purposes. It is common practise in content-based image retrieval systems to use multi-dimensional feature vectors to characterise the retrieved visual contents of the images in the database. The feature database is comprised of the feature vectors of all the photos in the database. Users feed the image retrieval system with reference photos or rough sketches to help it locate relevant results. Next, the system converts the examples into a feature vector form. Following this, an indexing strategy is used to determine the degree of similarity. Comparison is made between the feature vectors of the query example or sketch and the feature vectors of the images in the database, and a retrieval is made based on the result. This indexing method facilitates rapid image retrieval from a storage facility housing numerous photos. These days, users' relevant feedback is used to fine-tune the retrieval process and produce more perceptually and semantically correct, meaningful retrieval results.

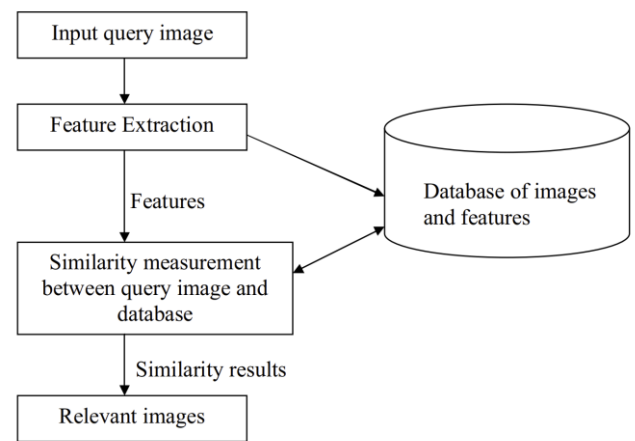


Fig. 3 Proposed process of digital image retrieval process

A wide variety of methods that are all at the cutting edge of their fields were chosen. All the tools and data sets were also supposed to be accessible from the internet and simple to put into action. Due to the high cost of experimentation, the image retrieval technique of similarity search was not explored. In addition, feature extraction was prioritised over other classification processes, such as pre-searching and user input.

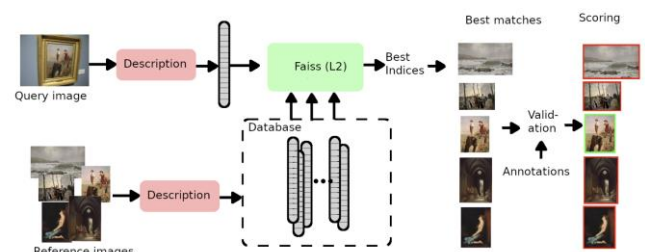


Fig. 4 Process of deeplearning based Image retrieval

To put the methods to the test, we first had to divide the dataset into a query database and a reference database. Photos were input in one at a time from both databases into most of the algorithms (with the exception of Votree and

R2D2), with the result being an N-dimensional vector characterising the images. In addition, this produced matrices with dimensions MN and LN, where M and L are the respective numbers of photos in the reference and query sets. Faiss-vector library's comparison system [8] was used to analyse the matrices. According to the L2-distances of the feature vectors, the best matches between the query photos and the reference images were revealed. (Shown in Fig. Only Votree, an open-source application employing SIFT features, and R2D2, which employed a custom-built brute-force strategy, stood out from the crowd.

The R2D2 [9, 10] system is a local feature detector and descriptor that uses deep learning. While similar methods, such as SIFT's keypoint detector and subsequent description, are commonplace, R2D2 does both. Once one feature tensor has been generated by the convolutional network, it is split and passed along to another set of networks. The first generates a pixel-level descriptor, while the second assesses and, ultimately, extracts keypoints from the image. Repeatability and reliability are calculated and evaluated independently from one another during extraction. Whereas repeatability indicates the likelihood that a given keypoint will be located in multiple images, reliability reveals how distinctive and evocative that point is. A model pre-trained on scene localization datasets was already included in the method, but an image retrieval module had to be developed separately. Both an exact clone of the Votree approach and a straightforward BOVW scheme with 1024 centroids estimated with k-means were tried, but neither produced satisfactory outcomes. R2D2 was deemed a crucial part of the experiment because it was the only locally-based deep learning approach that could be accessed on the internet. For this reason, we used R2D2's brute-force approach to image retrieval, which involved extracting keypoints and descriptors from each image, comparing each query image's descriptors with those of each database image, selecting the sum of L2-distances of 20 closest descriptor pairs per image, retrieving the database image with the smallest sum as the first result, etc. The results were equivalent using this brute-force approach, but it is obvious why this approach cannot be used in practise. This was a very costly process that took far longer than alternative search strategies. As a result, R2D2 only had to complete a fraction of the original 500 random samples for the COCO17 and SMVS-A tasks. The small sample size (500 samples) of SMVS-B necessitated using an unredacted data set. We further reduced the processing load by selecting a random subset of 100 records from the database that included the correct answer for each search.

IV. RESULTS

The confusion matrix is computed based on the order of the retrieved images and is displayed in Table 3.26. All accuracy values are computed using the confusion matrix, and the results are given in Table 1. The final acquired accuracy rate is 98%. Measures of quality like as precision, recall, specificity, and F1 score can be derived from the confusion matrix. All of the previous approaches apply the same treatment to the query image (image 270). Relevance images are obtained based on N, and from them, the error rate and NARR values are determined.

Table I: Predicted results for proposed method

	CLASS	PREDICTED			
		A	B	C	Total
ACTUAL	A	99	0	1	100
	B	3	97	0	100
	C	0	2	98	100
	Total	102	99	99	300

Table II: Classification results for parameters

	Accuracy	Precision	Recall	Specificity	F1 score
A	0.99	0.97	0.99	0.98	0.98
B	0.98	0.97	0.97	0.99	0.97
C	0.97	0.98	0.98	0.99	0.98
Over all	0.98	0.98	0.98	0.99	0.97

In Figure 5-9 we can see a comparison of the various approaches' accuracy, precision, recall, specificity, and F1 score values. It can be seen that the proposed method has the highest values for accuracy, precision, recall, specificity, and the F1 score. These numbers should ideally add up to 1 or 100%. Also, the proposed method has the lowest error rate and highest NARR values, making it the clear winner.

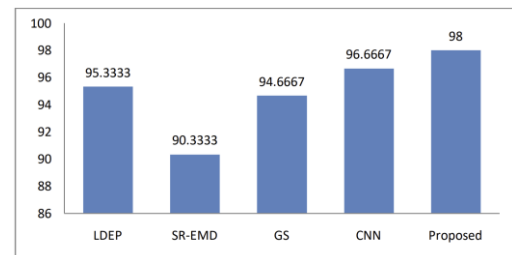


Fig. 5 comparison graph for Precision

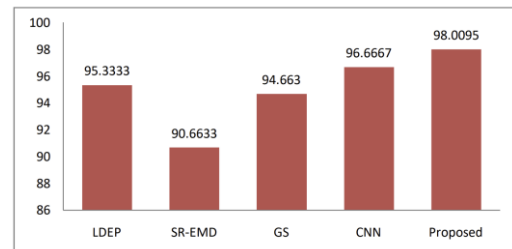


Fig. 6 Comparison graph for Accuracy

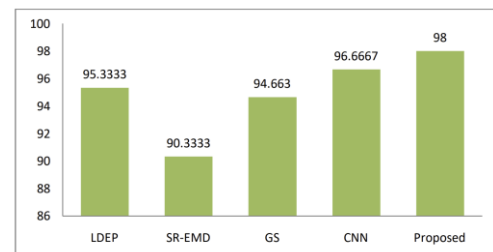


Fig. 7 comparison graph for recall

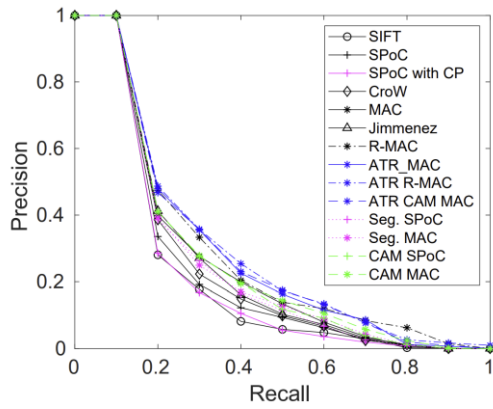


Fig. 8 Comparison graph for ROC curve

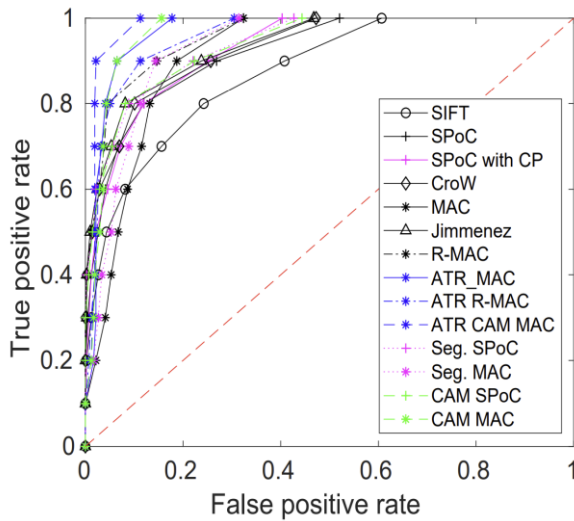


Fig. 9 Comparison of False positive and True Positive rate

CONCLUSION

We have presented component-based hard and soft attention solutions for boosting the performance of a DCNN based large-scale trademark retrieval (LSTR) system. Our hard attention based approaches for LSTR have achieved new state-of-the-art results, outperforming previous works by a significant margin. The experiments are conducted over different types of image retrieval problems. Focus was on feasibility and recentness of methods: Used codes were supposed to be available online, functional and run on a modest computing platform with only one GPU. Used datasets reflect the spectrum between visual and semantic sides of the problem. With the experiments and their results We have identified three unique use cases for picture retrieval and identified strategies that are optimal for each. Differences between global and local characteristics, as well as between classical and machine learning approaches, have also been discovered.

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